

Field Site & Objective

Location: Klein See, Cottbus

Setting: Wetland area: peat soils, high organic matter content, shallow water table.

Why here? High subsurface variability in moisture and organic content ... it is an ideal target for geophysical surveys

Objective: Map the subsurface structure to understand the spatial distribution of organic material, soil layers, and water saturation



Methods & Field Overview

Two methods applied along the same transect.

ERT: electrodes inserted in the ground, maps electrical resistivity of the subsurface.

EM: no electrodes needed, measures electromagnetic conductivity rapidly over large areas.

Both methods are non-destructive: no excavation, no soil disturbance

Complementary approach: ERT gives high vertical resolution; EM34 covers more area quickly

Instruments used:



a. Syscal Pro switch 48 meter



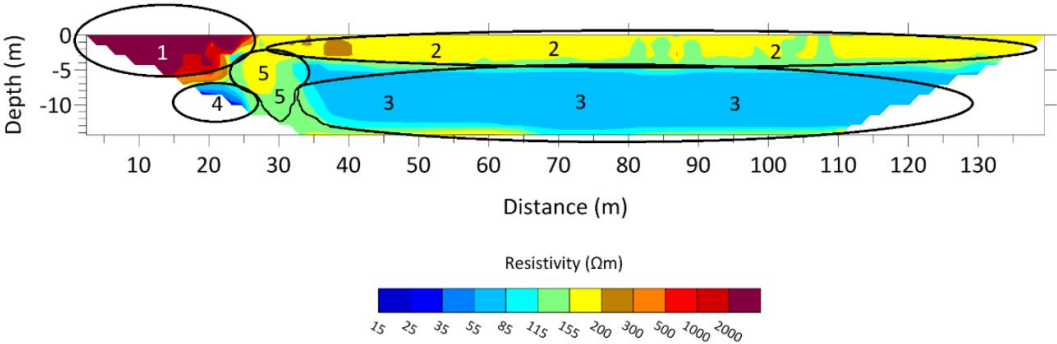
b. Electrodes



c. Connecting box



d. Cables



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Instruments used:



a. Electromagnetic coil



b. Receiver



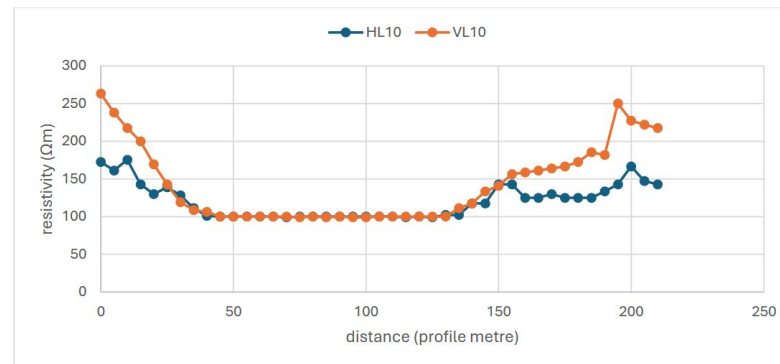
c. Transmitter



a. HLEM



b. VLEM



ERT vs EM

Pros & Cons summary:

- **ERT** → more detail, more control over depth, but slow and needs good electrode contact

- **EM** → fast, non-invasive, great for reconnaissance, but less vertical resolution and fixed depth ranges

Key:

- Used together: EM for quick spatial overview, ERT for detailed cross-section

	ERT	EM
Principle	Injects electrical current via electrodes	Electromagnetic induction, no ground contact
Measures	Resistivity (Ωm)	Conductivity (mS/m)
Electrodes	Yes: must be inserted in ground	No
Speed	Slow: setup takes time	Fast: walk along transect
Depth control	High: adjusted by electrode spacing	Limited: fixed by coil separation
Vertical resolution	High	Moderate
Lateral coverage	Limited	Large areas quickly
Difficult terrain	Hard (rocky, frozen ground)	No problem
Data quality	High detail	Good overview